SQL Date Functions and Date Math

1.
```
SELECT
        Order_id,
        Customer_id,
        order_date,
        DAYNAME (order_date) AS day_name
FROM orders;
```

2.
```
SELECT
        customer_id,
        customer_name,
        signup_date
        MONTHNAME(signup_date) AS signup_month_name
FROM customer_signups;
```

4.
```
SELECT
        transaction_id,
        customer_id,
        transaction_date,
        YEAR (transaction_date) AS transaction_year
FROM transactions;
```

3.
```
SELECT
        Sale_id,
        Product_name,
        sale_date,
        MONTH (sale_date) AS sale_month
FROM sales
```

5.
```
SELECT
        delivery_id,
        Customer_id,
        delivery_date,
        DAY(delivery_date) AS day_of_month
FROM deliveries
```

6.
```
SELECT
        employee_id,
        employee_name,
        department,
        CURRENT_DATE ( ) AS today_date
FROM employees;
```

7.
```
SELECT
        Order_id,
        customer_id,
        order_date_text,
        TO_DATE(order_date_text, 'YYYY-MM-DD') AS order_date
FROM online_orders;
```

8.
```
SELECT
        payment_id,
        customer_id,
        payment_date,
        TO_CHAR(payment_date, 'YYYY-MM-DD') AS formatted_payment_date
FROM payment_dates;
```

9. SELECT
```
        customer_id,
        customer_name,
        last_purchase_date,
        DATEDIFF(CURRENT_DATE(), last_purchase_date)
         AS days_since_last_purchase
 FROM customer_purchases;
```

10. SELECT
```
        order_id,
        customer_id,
        order_date,
        DATEADD(day, 7, order_date)
         AS expected_delivery_date
 FROM shipping_orders;
```

11. SELECT
```
        booking_id,
        customer_id,
        booking_date,
        YEAR(booking_date) AS booking_year,
        MONTH(booking_date) AS booking_month,
        DAY(booking_date) AS booking_date
 FROM bookings;
```

12. SELECT
```
        order_id,
        customer_id,
        order_date,
        YEAR(order_date) AS order_year,
        amount
FROM yearly_orders
WHERE YEAR(order_date) = 2026;
```

13. SELECT
```
        order_id,
        customer_id,
        order_date,
        MONTH(order_date) AS order_month,
        amount
 FROM monthly_orders
 WHERE MONTH(order_date) = 3;
```

14. SELECT
```
        subscription_id,
        customer_id,
        start_date,
        LAST_DAY(start_date) AS month_end_date
FROM subscriptions;
```

15. SELECT
```
        send_id,
        customer_id,
        send_date,
        DATE_TRUNC('MONTH', send_date)
         AS month_start_date
 FROM campaign_sends;
```

16. SELECT
```
        invoice_id,
        customer_id,
        invoice_date,
        TO_CHAR(invoice_date, 'Month YYYY')
 FROM invoices_dates;
```

```
17. SELECT
        customer_id,
        customer_name,
        date_of_birth,
        YEAR (CURRENT_DATE ()) - YEAR (date_of_birth)
          AS customer_age
  FROM customer_birthdays;
```

```
18. SELECT
        order_id,
        Customer_id,
        Order_date,
        DAYNAME (order_date) AS day_name,
        CASE
           WHEN DAYNAME (order_date) IN ('saturday', 'Sunday')
              THEN 'Weekend'
           ELSE 'Weekday'
        END AS day_type
FROM weekend_orders;
```

```
19. SELECT
        transaction_id,
        Customer_id,
        transaction_date,
        QUARTER (transaction_date)
          AS transaction_quarter,
        amount
FROM quarterly_transactions;
```

```
20. SELECT
        order_id,
        Customer_id,
        order_date,
        DATEDIFF (CURRENT_DATE (), order_date)
           AS days_since_order,
        amount
    FROM recent_orders
    WHERE DATEDIFF (CURRENT_DATE (), order_date) > 30;
```

```
Bonus Challenge. SELECT
        customer_id,
        customer_name,
        last_purchase_date,
        DATEDIFF (CURRENT_DATE (), last_purchase_date)
           AS days_since_last_purchase,
        CASE
           WHEN DATEDIFF (CURRENT_DATE (), last_purchase_date) <= 30
              THEN 'Active customer'
           WHEN DATEDIFF (CURRENT_DATE (), last_purchase_date)
              BETWEEN 31 AND 90
              THEN 'AT Risk Customer'
           ELSE 'Inactive Customer'
        END AS Customer_status
    FROM Customer_recency;
```